MediBook AI • Clinical Healthcare Platform

Intelligent AI Clinical Triage & Doctor Appointment System

Bridging patient symptoms to verified specialist care in under 45 seconds with guaranteed zero double-booking, real-time emergency triage, and DPDP Act 2023 compliance.

`Next.js 14 App Router` `Prisma ORM` `Redis Mutex Lock` `India DPDP Act 2023` `English • தமிழ் • తెలుగు`

Presentation Deck: Academic, Clinical & Technical Defense

Production Ready: 135 Doctors • 34 Chennai Hospitals • 20,790 Slots

PROBLEM ANALYSIS

Critical Challenges in Healthcare Scheduling

Clinical & Operational Gaps

Patient Misrouting

Patients lack medical knowledge and self-diagnose inaccurately (e.g., consulting cardiology for GERD or GP for neurological symptoms), delaying vital care.

DIAGNOSTIC DELAY

Peak Double-Booking

Traditional web forms fail to handle race conditions during high morning rush hours. Multiple users booking the exact same second causes clinic chaos.

CONCURRENCY FLAW

20–30% No-Show Rates

Rigid cancellation procedures and lack of instant rescheduling lead to unattended clinic slots and huge hospital revenue losses.

REVENUE WASTE

Language & A11y Gap

Elderly patients and non-English regional speakers face complex English jargon and tiny fonts, preventing independent digital adoption.

DIGITAL DIVIDE

Presenter Note: *Emphasize that traditional appointment portals are simply digital calendar forms without any clinical awareness, leading to misdiagnosis, slot conflicts, and accessibility failures.*

SOLUTION OVERVIEW

MediBook AI – The Smart Clinical Front Door

Core Value Proposition

What MediBook AI Delivers

- **AI-Guided Clinical Triage**: Evaluates unstructured patient symptoms in natural language (English / தமிழ் / తెలుగు) and maps to 14 verified medical specialties.
- **Traffic-Light Urgency Guardrails**: Immediate red-flag detection (crushing chest pain, stroke signs) that halts routine booking and surfaces 112 emergency escalation.
- **Zero Double-Booking Engine**: Atomic distributed lock holds slots for 180 seconds during patient review, guaranteed by composite database constraints.
- **Frictionless 3-Step Wizard**: Specialty ➔ Date & Slot with 180s Countdown ➔ Instant Confirmation with 1-click Calendar ICS download.

Conventional Portals vs. MediBook AI

Capability	Standard Portals	MediBook AI
Specialty Selection	Patient Guesswork	**AI Clinical Triage (45s)**
Double Booking	Frequent under rush	**Guaranteed 0% (Atomic Locks)**
Emergency Safeguards	None (Books regular)	**Instant RED Escalation (112)**
Languages & Senior UI	English Only	**Trilingual + A11y Font Scaler**
Cancellation Policy	Call front desk	**1-Click + 5s Undo Toast**

SYSTEM ENGINEERING

End-to-End System Architecture

Modern Next.js 14 Stack



1. Presentation Layer

- **Next.js 14 App Router** with React Server Components (RSC) for sub-second page loads.
- **Tailwind CSS 3.4** with full Dark & Light mode theme persistence.
- **i18n Context** for seamless English, Tamil, and Telugu translation switching.
- **A11y Font Scaler** (`A-` / `A` / `A+`) supporting senior patient accessibility.



2. Clinical & Concurrency Engine

- **Claude 3.5 & Gemini Integration** with offline deterministic clinical rule fallback.
- **Contextual Clarifier**: Triggers 1 targeted follow-up question when symptoms cross specialties.
- **Distributed Redis Mutex (`SET NX EX 180`)** holding slot reservations during checkout.
- **Atomic In-Memory Fallback** ensuring 100% reliability in standalone setups.



3. Data & Security Layer

- **Prisma ORM 5.22** with SQLite / PostgreSQL compatibility.
- **Physical Composite Constraint**: `@@unique([doctorId, startTime])` rejecting duplicates at the disk level.
- **AES-256 Encryption** for clinical consultation notes at rest.
- **DPDP Act 2023** Right to Erasure cascade deleting all patient records.

> ***Presenter Note:*** *Highlight the separation of concerns: Server Components render fast static views, while atomic Redis locks prevent concurrent checkout collisions before touching the database.*

AI Clinical Triage Protocol

ROUTINE TIER

GREEN Protocol

Criteria: Chronic conditions, mild cough/cold, general follow-ups, minor joint aches.

- Directly displays top-rated doctors sorted by specialty and hospital distance.
- Suggests self-care hydration and standard clinic appointment slots.

URGENT TIER

AMBER Protocol

Criteria: High fever (>102°F), acute abdominal pain, suspected fractures, severe ear infections.

- Promotes **Same-Day Priority Slots** within the next 4 hours.
- Pre-fills clinical urgency summary for the attending doctor's queue.

EMERGENCY STOP

RED Protocol

Criteria: Radiating chest pain, sudden slurred speech, unconsciousness, severe hemorrhaging.

- **Immediately halts booking wizard**.
- Flashes emergency alert & renders 1-click **Call 112** button.
- Displays nearest 24/7 trauma centers & ICU facilities.

Targeted Multi-Turn Clarification: When patient complaints overlap (e.g., "headache with blurry vision"), the AI does not guess—it triggers one precise clinical question ("Did the vision change happen suddenly, or have you had fever for days?") to distinguish Neurology from Ophthalmology.

CORE FEATURE 2

Zero Double-Booking Concurrency Engine

Dual-Layer Concurrency Architecture

Dual-Layer Concurrency Defense

1. **Layer 1: Distributed Redis Mutex (`SET NX EX 180`)**
   When a user clicks a slot, it is locked immediately for 180 seconds. A real-time countdown hold prevents any other user from selecting it during review.

2. **Layer 2: Database Unique Composite Constraint**
   At commit time, the schema enforces `@@unique([doctorId, startTime])`. Any concurrent collision is rejected at the database engine level with zero data corruption.

Automated Concurrency Stress Test Results

50	0	100%
PARALLEL REQUESTS	DOUBLE BOOKINGS	DATA INTEGRITY

Test Suite Verification (npm run test:concurrency):

- Concurrency test: 50 simultaneous checkout calls ➔ 1 succeeded, 49 rejected gracefully.
- AI Triage schema validation test ➔ Passed 100%.
- Emergency RED tier escalation test ➔ Passed 100%.
- Atomic slot release upon cancellation test ➔ Passed 100%.

ECOSYSTEM SCALE

Production Dataset – Chennai Healthcare Ecosystem

Real-World Scale

135
VERIFIED DOCTORS

34
PREMIER HOSPITALS

20,790
ACTIVE TIME SLOTS

14
CLINICAL SPECIALTIES

Top Partner Hospitals

- **Apollo Hospitals** (Greams Road & OMR) – Cardiology & Oncology centers
- **Fortis Malar Hospital** (Adyar) – Advanced Cardiac & Critical Care
- **MIOT International** (Manapakkam) – Orthopedics & Trauma surgery
- **Dr. Rela Institute** (Chromepet) – Hepatobiliary & Multi-organ care
- **Gleneagles Health City** (Perumbakkam) – Multispecialty tertiary care

Specialty Distribution

Medical Coverage: Cardiology, Neurology, Orthopedics, Pediatrics, Dermatology, General Medicine, Gastroenterology, Pulmonology, Nephrology, Ophthalmology, ENT, Gynecology, Psychiatry, and Oncology.

Realistic Profiles: Verified qualifications (MBBS, MD, DM, FRCS), 7 to 36 years experience, genuine Chennai addresses, landmarks, consultation fees (₹500 - ₹1200), and English + Tamil bios.

USER EXPERIENCE

Multi-Role Portals (Patient, Doctor, Admin)

Role-Based Access

Patient Experience Hub

- **3-Step Booking Wizard**: Smooth stepper flow with 180s countdown hold.
- **Self-Service Rescheduling**: 1-click date & slot modification.
- **Safe Cancellation**: Confirmation modal with a **5-second Undo Toast** safeguard.
- **Add to Calendar**: Instant `.ics` calendar file generation with built-in alarm alerts.

Doctor Consultation Portal

- **Live Queue Management**: View today's registered patients sorted chronologically.
- **AI Clinical Pre-Brief**: Symptom summaries and triage level ready before consultation.
- **Queue Controls**: 1-click status buttons (In-Consultation, Completed, No-Show).
- **Emergency Slot Blocker**: 1-click freeze to preserve slots for walk-in ICU emergencies.

Admin Analytics Dashboard

- **Operational KPIs**: Total bookings, completed visits, active doctors, and revenue.
- **Real-Time No-Show Rate (%)**: Identifies attendance bottlenecks across clinics.
- **Specialty Demand Heatmap**: Analyzes high-demand medical departments.
- **Doctor Capacity Utilization**: Monitors slot occupancy across shifts.

COMPLIANCE & GOVERNANCE

Security & India DPDP Act 2023 Compliance

Privacy by Design

AES-256 Data Encryption

- Clinical consultation notes, symptom logs, and patient records are encrypted at rest using industry-standard **AES-256-GCM**.
- Cryptographic keys stored securely via environment variables with zero plaintext database exposure.

Zero PII to AI Models

- Patient PII (names, phone numbers, email addresses, IP addresses) is **strictly sanitized** prior to clinical triage calls.
- AI evaluation processes only sanitized symptom text and duration metadata, preventing unauthorized data mining.

DPDP Act 2023 Principles

- **Informed Consent**: Explicit consent banner before symptom submission.
- **Purpose Limitation**: Data used strictly for appointment scheduling.
- **Right to Erasure ("Delete My Data")**: 1-click button in profile permanently purging all personal records from the DB.

Presenter Note: *Emphasize that healthcare applications in India must adhere to the Digital Personal Data Protection (DPDP) Act of 2023. MediBook AI satisfies this by implementing data minimization, anonymization, and the Right to Erasure.*

INCLUSIVITY & ACCESS

Accessibility & Regional Language Support

Universal Healthcare Access

Trilingual Localization

Seamless one-click language toggle across all patient-facing workflows:

- **English**: Global clinical terminology.
- **தமிழ் (Tamil)**: Full regional translation covering symptoms, specialty names, and doctor bios.
- **తెలుగు (Telugu)**: Complete localized interface for regional accessibility.

Senior Citizen Font Scaler

Dedicated typography scaling controls positioned permanently in the navigation header:

- **A-**: Standard compact density (100%).
- **A**: Default readability mode (110%).
- **A+**: High-visibility senior mode (125%) with persistent local storage saving.

WCAG 2.1 AA Compliance

Adheres to global web accessibility guidelines:

- 4.5:1 minimum contrast ratio across all text and buttons.
- Full keyboard navigation (Tab, Enter, Escape).
- Screen-reader accessible ARIA labels for countdown timers and modals.

CLOUD ENGINEERING

Technology Stack & Cloud Deployment

Production Architecture

Full Stack Specification

Framework	Next.js 14.2.15 (App Router, Server Actions, API Routes)
Language	TypeScript 5.6 (Strict Type Safety, 0 build warnings)
Styling	Tailwind CSS 3.4 + Lucide React Icons
ORM & Database	Prisma 5.22 with SQLite / PostgreSQL
Concurrency	Redis Mutex (ioredis) + Atomic In-Memory Fallback
AI Engine	Claude 3.5 / Gemini API + Deterministic Clinical Router

Deployment & DevOps

- **Netlify Cloud Infrastructure**: Deployed with `@netlify/plugin-nextjs` runtime v5 providing global CDN delivery and automatic edge caching.
- **Zero Downtime Serverless DB**: Bundled SQLite dev database auto-copied to `/tmp` in serverless lambdas to prevent read-only filesystem locks.
- **Automated CI/CD Pipeline**: Pushes to GitHub `main` trigger automated builds, type validation, and production deployments.
- **100% Self-Contained Repository**: Clone and run `npm run dev` with full seeded database ready out of the box.

IMPACT & VISION

Measurable Impact, Roadmap & Conclusion

Future Healthcare Vision

Demonstrated Project Impact

- **70% Faster Scheduling**: Reduces appointment time from 4 minutes of confusing form-filling to under 45 seconds of natural language triage.
- **Zero Booking Conflicts**: 100% elimination of double-booked clinic slots under heavy concurrent loads.
- **Reduced ER Congestion**: Prevents non-urgent visits to trauma centers while guaranteeing immediate 112 escalation for true emergencies.
- **Maximized Clinic Utilization**: Real-time slot recycling on cancellation recovers open slots immediately.

Strategic Roadmap

- **ABDM (Ayushman Bharat) Integration**: Link verified ABHA health ID numbers for automated health record exchange.
- **Automated WhatsApp Notifications**: Send interactive appointment reminders and 1-click confirmation buttons to cut no-show rates below 5%.
- **Tele-Health WebRTC Rooms**: Integrated high-definition video consultations for remote rural patients.

"MediBook AI: Transforming medical appointments into an intelligent, accessible, and conflict-free clinical front door."